

Smart Document Similarity and Plagiarism Detection System

Review 2 — Data Structures & Algorithms | DSA-3 Project

Team Members

Vidya Keerthi Chitturi — 2510030372

Shaik Asha Parveen — 2510030225

Madi Amulya — 2510030158

Badam Hasika — 2510030403

Project Details

Section: A7

Group: G-9

Introduction

The Smart Document Similarity and Plagiarism Detection System is a **proposed system** for comparing documents and identifying exact, modified, and common content between them. The system will preprocess document text and apply efficient Data Structures and Algorithms to detect matching patterns, measure similarity, and generate a detailed comparison report. The project's core focus is applying advanced string algorithms from DSA-3 to a real-world document-analysis problem.

What the System Will Do

- Preprocess and tokenize raw document text for algorithmic comparison
- Detect exact, near-duplicate, and structurally similar content across documents
- Generate a comprehensive similarity report with matched regions highlighted

Motivation

- Plagiarism can involve direct copying as well as modified or rephrased content
- Manual comparison of large documents is time-consuming and error-prone
- Existing research spans textual similarity, hybrid methods, ML, and semantic analysis
- Our project explores a focused, **algorithm-driven DSA approach** to document similarity

Paper 1: Hybrid Weighted Similarity

Improving Plagiarism Detection in Text Document Using Hybrid Weighted Similarity

Authors: Hamed Arabi and Mehdi Akbari | **Year:** 2022 | **Journal:** Expert Systems with Applications

Problem

Plagiarism detection must handle not just direct copying but also modified and rephrased content.

Approach

Hybrid weighted similarity method with multi-stage filtering at document and sentence levels.

Key Finding

Multi-stage similarity approach is more effective than single-method matching for detecting intelligent plagiarism.

Limitation

The hybrid framework adds complexity compared to basic textual matching approaches.

Paper 2: Feature Selection + SVM

An Effective Text Plagiarism Detection System Based on Feature Selection and SVM Techniques

Authors: Mohamed A. El-Rashidy, Ramy G. Mohamed, Nawal A. El-Fishawy and Marwa A. Shouman | **Year:** 2023 | **Journal:** Multimedia Tools and Applications

Study Overview

- **Problem:** Plagiarism detection must handle lexical, syntactic, and semantic variations in text
- **Approach:** Uses feature selection to identify effective similarity metrics, then applies **SVM** classification to distinguish similarity cases
- **Key Finding:** Feature selection combined with SVM can effectively classify multiple types of plagiarism beyond direct copying
- **Evaluation:** Tested on **PAN 2013** and **PAN 2014** benchmark datasets

Assessment

Strength

Handles multiple forms of text plagiarism — not just direct copying — by learning from labeled similarity features across lexical, syntactic, and semantic dimensions.

Limitation

Requires machine learning infrastructure and feature engineering, adding complexity beyond pure algorithmic string matching.

Paper 3: Systematic Review of Techniques

Comparative Analysis of Text-Based Plagiarism Detection Techniques

Authors: Muhammad Sajid, Muhammad Sanaullah, Muhammad Fuzail, Tauqeer Safdar Malik and Shuhaida Mohamed Shuhidan | **Year:** 2025 |

Journal: PLOS ONE

Problem

Plagiarism detection remains challenging, especially for modified, paraphrased, and obscured content.

Approach

Systematic review of **189 research papers** (2019–2024) examining traditional textual approaches, semantic analysis, machine learning, and other computational methods.

Key Finding

No single approach solves all plagiarism detection cases; combining multiple methodologies is important for future work.

Limitation

Even advanced approaches struggle with highly obscured plagiarism, indicating the problem remains open.

Analysis of Existing Approaches

Study	Approach	Strength	Limitation
Arabi & Akbari (2022)	Hybrid weighted similarity	Handles modified content	More complex than basic matching
El-Rashidy et al. (2023)	Feature selection + SVM	Handles multiple plagiarism types	Requires ML infrastructure
Sajid et al. (2025)	Systematic review of techniques	Broad overview of field	Shows problem remains open

Key Observations

- Simple direct matching is insufficient for all plagiarism cases
- Advanced approaches improve detection but add computational or model complexity
- No universal solution exists; different techniques address different aspects
- Efficient text-pattern algorithms remain relevant for exact and structural similarity

Research Gap & Proposed DSA Pipeline

Research Gap

Based on the literature analysis, we propose a focused DSA-based approach that combines advanced string algorithms to detect different forms of document similarity efficiently.

This pipeline emphasizes exact, fuzzy, and substring-based analysis without relying on ML-based classification or semantic models.

Technical Workflow

01	02	03
Document Upload	Text Extraction & Preprocessing	Exact Pattern Matching
		Naïve KMP Z-Algorithm Rabin-Kar
04	05	06
Fuzzy Matching	Common Substring Analysis	Similarity Analysis
Edit Distance / Dynamic Programming	Suffix Array + LCP / Kasai	
07		

Similarity Report

DSA-3 supporting concepts: arrays/lists, hash maps, searching, and sorting underpin all pipeline stages as foundational data management infrastructure.

Core Algorithms in the Proposed System

The proposed pipeline is built around **DSA-3 core algorithms** for string analysis. Each algorithm group addresses a specific detection task in the document comparison workflow.

Exact Pattern Matching

- **Naïve Algorithm:** Brute-force pattern search
- **KMP (Knuth-Morris-Pratt):** Efficient pattern matching with failure function
- **Z-Algorithm:** Linear-time pattern matching using Z-array
- **Rabin-Karp:** Hash-based pattern matching for multiple patterns

Purpose: Detect exact copies of words, phrases, and sentences.

Fuzzy Matching

- **Edit Distance (Levenshtein):** Measure similarity between strings
- **Dynamic Programming:** Compute minimum edits needed to transform one string to another

Purpose: Detect modified and near-duplicate text with character-level variations.

Common Substring Analysis

- **Suffix Array:** Efficient data structure for substring queries
- **LCP Array (Kasai Algorithm):** Compute longest common prefixes

Purpose: Identify long common sections and structural similarities between documents.

PLANNED FEATURES

Planned System Features

Detection Capabilities

- Exact matching of copied words, phrases, and sentences
- Detection of modified and near-duplicate text using Edit Distance
- Identification of long common sections using suffix structures
- Efficient pattern searching using advanced string algorithms

Analysis & Reporting

- Highlighting of matching content in comparison view
- Comparison of multiple documents simultaneously
- Similarity percentage and document ranking
- Detailed similarity report with pattern analysis

Conclusion

The literature review confirms that plagiarism detection remains an active and challenging problem — particularly when content is modified, paraphrased, or otherwise obscured. Based on these observations, our project proposes a focused, algorithm-driven solution.

What the Literature Shows

- Hybrid similarity methods, ML, and semantic analysis each contribute to detection but introduce model or infrastructure complexity
- Detecting modified and obscured plagiarism remains unsolved even with advanced approaches
- Efficient text-pattern algorithms remain a relevant and underexplored standalone direction

Our Proposed Direction

- Combine **Naïve / KMP / Z-Algorithm / Rabin-Karp** for exact pattern matching
- Apply **Edit Distance (DP)** for fuzzy and near-duplicate detection
- Use **Suffix Array + LCP (Kasai)** for common-substring analysis
- Deliver a practical application of DSA-3 algorithms to real-world document analysis

The planned implementation will demonstrate how composing multiple advanced string algorithms into a structured pipeline can address different forms of document similarity without requiring machine learning infrastructure.